Circular Motion Kinematics

SYNOPSIS

I summarize the rotational kinematics equations. Note that rotational kinematics is just like linear kinematics. Instead of linear displacement, you have an angular displacement since the motion has circular symmetry, so the distance from the center of rotation is fixed, and only the angle subtended by the motion changes while rotating. There are equations for angular velocity and angular acceleration that look just like the equations for velocity and acceleration in linear kinematics.

The angular displacement gives the arc length or the actual distance covered, and the angular velocity gives the tangential velocity for a given radius of circular trajectory (see below). Angular displacement is θ, and the arc length is $s = r\theta$, *provided* θ is in radians. That proviso is why we bother with radians at all: in degrees the same statement needs a conversion factor bolted onto it, and you would carry that factor through every line of every derivation that follows. A full circle is 2π radians, and "one revolution," "one rotation" and "one cycle" all mean the same trip. Angular velocity is $\omega = \Delta\theta/\Delta t$ in rad/s, and the tangential velocity it gives you at radius r is $v_t = r\omega$. Notice that ω has no r in it and v_t does: every point on a rigid turning object has the same angular velocity, and a different speed.

That is as far as this Note goes. **Note 2** picks up the other half of the story — centripetal acceleration and the force that supplies it — and the later notes turn α back on.

BEFORE YOU READ ON — PREDICT

Two children sit on the same merry-go-round, one near the hub and one out at the rail. Are they moving at the same **speed**, and are they turning at the same **rate**? Commit to a sentence, in your guided notes. If your answer to the two halves is the same word, one of them is wrong.

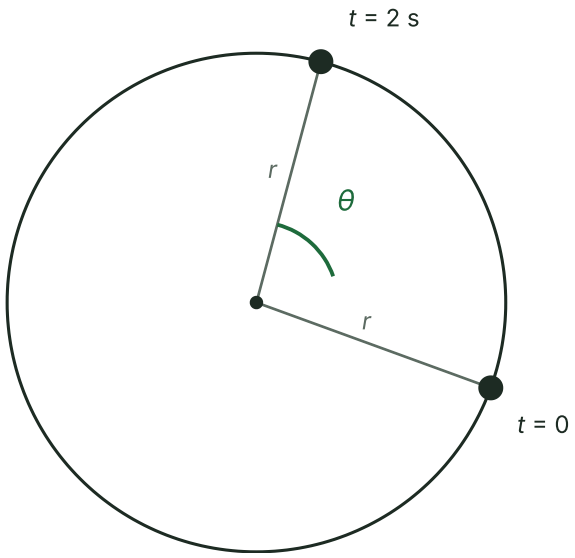
PERFORMANCE EXPECTATIONS

Apply rotational kinematics equations to solve circular and centripetal force problems.

For PHY 151 — with calculus: Define the angular variables as derivatives, $\omega = d\theta/dt$ and $\alpha = d\omega/dt$, and recover the three kinematics equations by integration rather than by analogy.

Angular displacement, and the radian

Fix a centre and a radius and the only thing left to say about a particle on that circle is its angle. Two things to keep straight: the arc is the distance actually travelled, so $s = r\theta$ is a real distance in metres; and that formula is true in radians and false in degrees. Degrees to radians is $\times 2\pi/360$, radians to degrees is $\times 360/2\pi$, and π rad = 180° is the one worth keeping in your head.

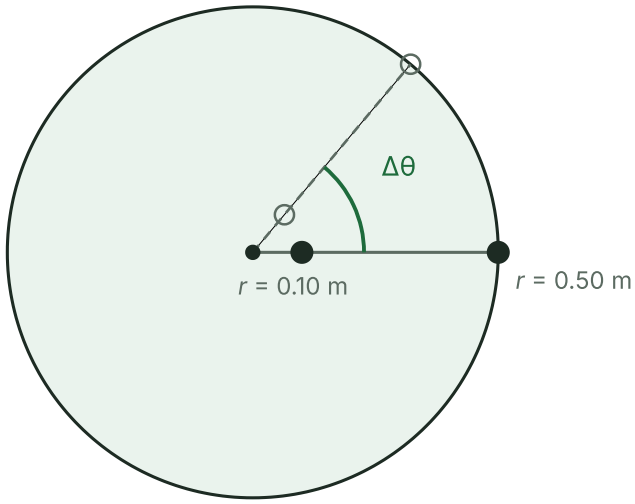


Mark the arc the object travelled between the two instants. At each instant draw the velocity, then the net force, and label the angle between them.

1. Express the following angles in radians: 10°, 50°, 75°, 260°.
2. The following angles are in units of radians. Express them in degrees: $\pi/3$, 0.55π, 1.8π, 4π.
3. Express those same four angles in units of revolutions.
4. A sector of radius 2.0 m subtends 75°. Find the arc length twice — once as $d = 2\pi r(\theta°/360)$, once as $s = r\theta$ — and say why they agree.

Angular velocity, and the tangential speed it gives you

$\omega = \Delta\theta/\Delta t$, in rad/s, and never in rev/min once you are inside an equation. Every rotating machine in every problem is quoted in rev/min, so the first line of the working is almost always the same one: multiply by $2\pi/60$. Then $v_t = r\omega$ turns that spin rate into a speed — a different speed at every radius, from the same ω.



Both riders sit on the same spoke of the same merry-go-round.

Δθ is the angle the spoke sweeps; the open circles are where each rider has got to after that same interval.

Draw each rider's velocity arrow, to scale with the radii as they are drawn here. Then mark the angle each one sweeps in the same interval.

5. Microwave ovens rotate at a rate of about 7 rev/min. (A) What is this in revolutions per second? (B) What is the angular velocity in radians per second?

6. A merry-go-round makes 2.5 rotations in 20 s. (A) What is its angular velocity? (B) What is the speed of a rider sitting 0.50 m from the centre? (C) And of a rider sitting 0.10 m from the centre?

7. In lacrosse, a ball is thrown from a net on the end of a stick by rotating the stick and forearm about the elbow. If the angular velocity of the ball about the elbow joint is 30.0 rad/s and the ball is 1.2

m from the elbow joint, what is the velocity of the ball?

8. From your answers to 6(B) and 6(C): which of ω and v_t depends on where you sit, and which does not? One sentence.

The rotational kinematics equations

Define $\alpha = \Delta\omega/\Delta t$, the angular acceleration, in rad/s². Now every constant-acceleration equation from Module 1 has a twin with θ in place of d — not an analogy, the same derivation in different letters. Fill the right-hand column before you reveal it. This is the table you will turn back to for the rest of the module.

9. Complete the right-hand column, including the five bridge rows that connect the angular quantities to the linear ones.
10. One of those bridge rows plays no part anywhere in this module. Which one, and why not?
11. What unit must ω be in for every row to work, and by what factor is an answer wrong if you feed in rev/min instead?

Linear kinematics	Rotational kinematics
$d = d_0 + vt + \frac{1}{2}at^2$	
$v = v_0 + at$	
$v^2 = v_0^2 + 2ad$	
arc length	
tangential velocity	

tangential acceleration	
centripetal acceleration	
centripetal force	

FOR PHY 151 — WITH CALCULUS

The ratios above are averages over Δt. The instantaneous quantities are the derivatives, exactly as in linear kinematics: $\omega = d\theta/dt$ and $\alpha = d\omega/dt = d^2\theta/dt^2$. Written that way, the three rotational kinematics equations are not an analogy you have to trust — they are what you get by integrating α twice, which is the Module 1 calculation in different letters.

12. Integrate α once and then twice to recover $\omega = \omega_0 + \alpha t$ and $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$, and say exactly where the assumption "α constant" is used.
13. For uniform circular motion the position vector is $\vec{r}(t) = r\,(\cos \omega t)\,\hat{\imath} + r\,(\sin \omega t)\,\hat{\jmath}$. Differentiate once and verify two things about the velocity: that it is perpendicular to $\vec{r}$ at every instant, and that its magnitude is $r\omega$. You have just proved the two statements the first figure asserted.

Looking ahead

You now have the variables and the kinematics: an angle in radians, a spin rate that belongs to the whole object, and a table of equations you already knew in different letters. **Note 2** asks the question this Note deliberately left alone — if the velocity is turning, there is an acceleration and therefore a force, so how big are they and which way do they point? That is centripetal acceleration and centripetal force, and it is where the last two rows of the table above get used. **Note 1a** is the in-class assignment that bridges the two.

Where this sits. Everything here is Module 1 in different letters: θ, ω and α are *d*, *v* and *a* with circular symmetry imposed, which is why the three kinematics equations came back unchanged. Module 3's $F = ma$ is what Note 2 brings to them.

BACK TO YOUR PREDICTION

Same rate, different speed. Both children sweep the same angle in the same time, so ω is identical — it is one rigid object. But $v_t = r\omega$, so the one at the rail is moving several times faster and needs several times the force to stay put. If you wrote "the same" for both halves you were answering about ω and forgot *r*; if you wrote "different" for both, you were answering about *v* and forgot that the merry-go-round is one rigid object.

Check yourself

1. A wheel turns at 45.0 rev/min. What is its angular velocity in rad/s?

_____ rad/s

2. Two riders sit on the same turning merry-go-round, one at 1.0 m from the hub and one at 3.0 m. Which statement is true?

- ○ Same tangential speed, and the outer rider has one third the angular velocity
- ○ Same angular velocity, and the outer rider has three times the tangential speed
- ○ Same angular velocity and the same tangential speed — it is one rigid object
- ○ The outer rider has three times the angular velocity and three times the tangential speed

3. The arc length formula $s = r\theta$ is

- ○ true for any unit of angle
- ○ true only when θ is in radians
- ○ true only when θ is in degrees
- ○ true only for a whole number of revolutions

4. A wheel of radius 0.30 m turns through 2.0 rad. How far does a point on its rim travel, in m?

_____ m

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then work the items as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 5 closes.

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